Few-Shot Visual Adaptation of a Pretrained ViZDoom Agent

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Motivation and Research Question

A pretrained CNN+GRU agent performs well in the original ViZDoom Deadly Corridor, but visual domain shifts can sharply reduce completion. We ask:

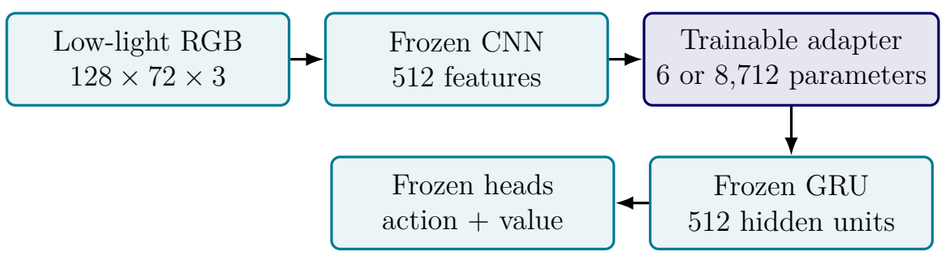
Can the agent adapt to low light from only a few observations without retraining its policy?

We reduce image brightness to 0.85 and adapt with matched clean/dark observations. Performance is measured on unseen episode seeds using completion rate.



Figure 1: A Call-of-Duty-style FPS context motivates robust perception under lighting shifts; experiments use ViZDoom Deadly Corridor.

Frozen-Policy Adapter Architecture



The post-CNN residual bottleneck is identity-initialized:

$$h' = h + W_{up}\,\sigma(W_{down}h).$$

All 22 original Hugging Face policy tensors remain bit-identical. Only the adapter learns from policy-logit and value consistency.

Controlled Experiments

1. Placement: six parameters before CNN vs. after CNN.
2. Capacity: six-parameter grouped affine vs. 8,712-parameter residual bottleneck.
3. Data budget: 128, 512, 1,024, or 2,048 observations (512–8,192 game frames).
4. Baselines: frozen upstream and full CNN+GRU fine-tuning.
5. Confirmation: frozen best checkpoint, 500 fresh low-light and 500 fresh clean episodes per policy.

Three independent adaptation seeds are used throughout, with matched evaluation seeds and paired outcomes.

Parameter-Matched Placement

Six-parameter adapter	Low-light mean	Worst seed	Clean mean
Before CNN (RGB)	58.67%	58%	–
After CNN (features)	68.00%	67%	84.33%
Frozen upstream	–	–	86.00%

At equal capacity and equal 8,192-frame budget, post-CNN placement won all three matched comparisons and lost only 1.67 clean points.

Few-Shot Budget Curve

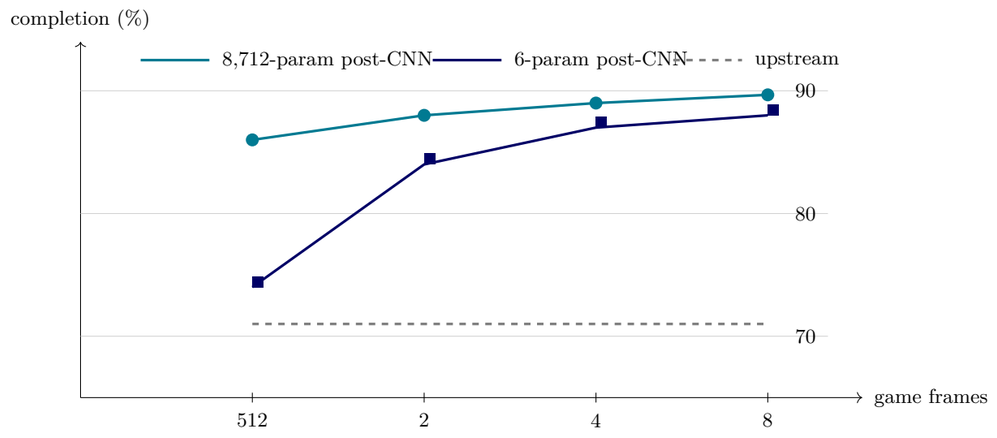


Figure 2: Low-light completion on 100 unseen episodes per policy. The larger post-CNN adapter reaches 86% after only 512 game frames.

Full CNN+GRU fine-tuning could not realize fewer than 8,192 frames and scored 68% there, below its frozen baseline (69%).

Final 500-Episode Confirmation

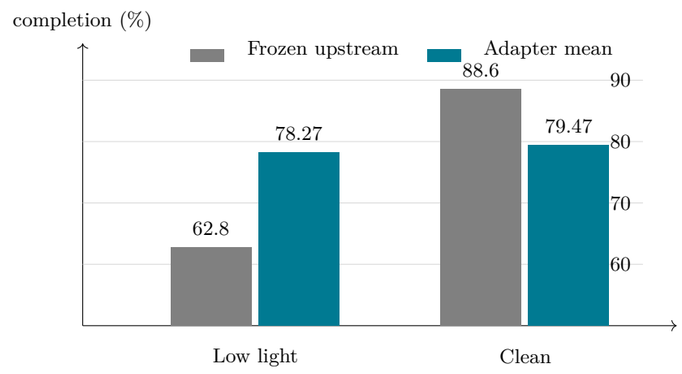


Figure 3: Mean completion across three independently adapted checkpoints on fresh, disjoint episode seeds.

Low light: confirmed
Upstream: 62.8%
Adapter: 78.27%
+15.47 points

Clean retention: failed
Upstream: 88.6%
Adapter: 79.47%
−9.13 points

All three adapter seeds won their paired low-light comparison. Candidate-only versus upstream-only successes were 96:11, 91:10, and 89:23.

What Worked—and What Did Not

- Strong specialist: 128 observations (512 frames) were sufficient for a reproducible low-light gain.
- Safe small adapter: the six-parameter post-CNN adapter reached 74% low-light and 88% clean at 512 frames.
- Capacity tradeoff: 8,712 parameters adapted faster, but exceeded the predefined five-point clean-retention limit.
- Full fine-tuning was inefficient: its smallest update required 16× more frames and did not improve low-light completion.
- Zero-shot generalization remained unsolved: multi-domain GRU and GRU+attention training scored only 3.33% and 2.8% on the sealed combined shift.

Conclusion

A frozen post-CNN adapter can specialize a recurrent ViZDoom policy to low light with only 512 game frames.

The 8,712-parameter adapter is best deployed as a low-light specialist behind a shift detector. The six-parameter post-CNN adapter is the safer always-on option because it improves low-light performance while preserving the clean domain.

Next step: learn a conditional gate that activates the specialist only under detected visual shift, combining high adaptation gain with clean-domain retention.

Reproducibility

ViZDoom Deadly Corridor; pretrained Hugging Face CNN+GRU; brightness 0.85; three adaptation seeds; disjoint matched test seeds; frozen-policy hash audits; no tuning on confirmation seeds.

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